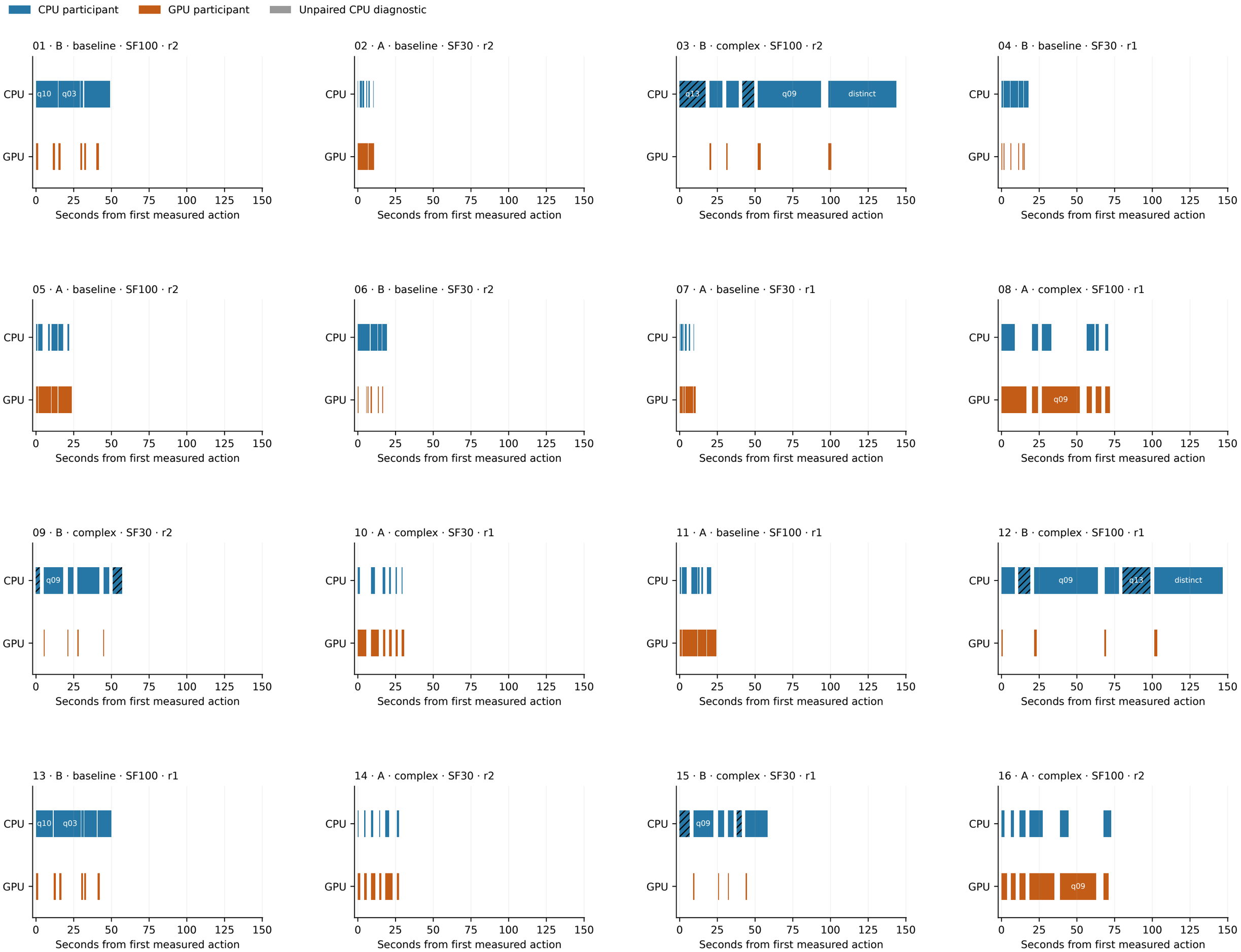
# Expansion · actual measured action intervals

**Separate descriptive extension · COMPLETE**




Every attempted expansion wave appears in planned execution order. Bars retain actual measured UTC start/end intervals and gaps, with a shared linear scale; fast bars are never widened. Zero is each wave's first measured action, not process startup. CPU/GPU lanes show concurrent participants; warmups are omitted. A: Spark GPU + CPU Polars 2.0.0. B: CPU Spark + Polars GPU. Hatched pair-B Q13/window CPU actions are unpaired diagnostics. Labels appear only where space permits. Both-clock common-query overlap is independently checked in the source audit; the timeline is not a process-lifetime proxy.